

SOCKET PROGRAMMING

Experiment 1 - Basic TCP Echo Client and Echo Server

Server Code – exp1_server.c

```
/* Name-Sumit Mund
   date-18-08-2026
   sic - 24bcsh05
*/

/* Experiment 1 : Implementation of a Basic TCP Echo Client and Echo Server
   SERVER : receives a message and echoes the same message back to the client */

#include <stdio.h>
#include <sys/types.h>
#include <sys/socket.h>
#include <netinet/in.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

void error(char *msg)
{
    perror(msg);
    exit(1);
}

int main(int argc, char *argv[])
{
    int sockfd, newsockfd, portno, clilen;
    char buffer[256];
    struct sockaddr_in serv_addr, cli_addr;
    int n;

    if (argc < 2) {
        fprintf(stderr, "ERROR, no port provided\n");
        exit(1);
    }

    sockfd = socket(AF_INET, SOCK_STREAM, 0);
    if (sockfd < 0)
        error("ERROR opening socket");

    bzero((char *) &serv_addr, sizeof(serv_addr));
    portno = atoi(argv[1]);
    serv_addr.sin_family = AF_INET;
    serv_addr.sin_addr.s_addr = INADDR_ANY;
    serv_addr.sin_port = htons(portno);

    if (bind(sockfd, (struct sockaddr *) &serv_addr, sizeof(serv_addr)) < 0)
        error("ERROR on binding");

    listen(sockfd, 5);

    clilen = sizeof(cli_addr);
    newsockfd = accept(sockfd, (struct sockaddr *) &cli_addr, &clilen);
    if (newsockfd < 0)
        error("ERROR on accept");
```

```
bzero(buffer, 256);
n = read(newsockfd, buffer, 255);
if (n < 0)
    error("ERROR reading from socket");

printf("Here is the message: %s\n", buffer);

n = write(newsockfd, buffer, strlen(buffer));
if (n < 0)
    error("ERROR writing to socket");

return 0;
}
```

Client Code – exp1_client.c

```
/* Name-Sumit Mund
   date-18-08-2026
   sic - 24bcsh05
*/

/* Experiment 1 : Implementation of a Basic TCP Echo Client and Echo Server
   CLIENT : sends a message to the server and prints the echoed message */

#include <stdio.h>
#include <sys/types.h>
#include <sys/socket.h>
#include <netinet/in.h>
#include <netdb.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>

void error(char *msg)
{
    perror(msg);
    exit(0);
}

int main(int argc, char *argv[])
{
    int sockfd, portno, n;
    struct sockaddr_in serv_addr;
    struct hostent *server;
    char buffer[256];

    if (argc < 3) {
        fprintf(stderr, "usage %s hostname port\n", argv[0]);
        exit(0);
    }

    portno = atoi(argv[2]);

    sockfd = socket(AF_INET, SOCK_STREAM, 0);
    if (sockfd < 0)
        error("ERROR opening socket");

    server = gethostbyname(argv[1]);
    if (server == NULL) {
        fprintf(stderr, "ERROR, no such host\n");
        exit(0);
    }
}
```

```
    bzero((char *) &serv_addr, sizeof(serv_addr));
    serv_addr.sin_family = AF_INET;
    bcopy((char *) server->h_addr, (char *) &serv_addr.sin_addr.s_addr, server-
>h_length);
    serv_addr.sin_port = htons(portno);

    if (connect(sockfd, (struct sockaddr *) &serv_addr, sizeof(serv_addr)) < 0)
        error("ERROR connecting");

    printf("Please Enter the Message: ");
    bzero(buffer, 256);
    fgets(buffer, 255, stdin);

    n = write(sockfd, buffer, strlen(buffer));
    if (n < 0)
        error("ERROR writing to socket");

    bzero(buffer, 256);
    n = read(sockfd, buffer, 255);
    if (n < 0)
        error("ERROR reading from socket");

    printf("Echo from server: %s\n", buffer);

    close(sockfd);
    return 0;
}
```

Compile & Run Commands

Terminal 1: ./expl_server 8100
Terminal 2: ./expl_client localhost 8100

Sample Interaction

Input used: Client typed: Hello World

Sample Output

Server Output:

Here is the message: Hello World

Client Output:

Please Enter the Message: Echo from server: Hello World

Experiment 2 - TCP Echo Server that repeats until the client types "Bye"

Server Code – exp2_server.c

```
/* Name-Sumit Mund
   date-24-08-2026
   sic - 24bcsh05
*/

/* Experiment 2 : TCP Echo Server that repeatedly echoes the message
   back to the client until the client types "Bye" -- SERVER */

#include <stdio.h>
#include <sys/types.h>
#include <sys/socket.h>
#include <netinet/in.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

void error(char *msg)
{
    perror(msg);
    exit(1);
}

int main(int argc, char *argv[])
{
    int sockfd, newsockfd, portno, clilen;
    char buffer[256];
    struct sockaddr_in serv_addr, cli_addr;
    int n;

    if (argc < 2) {
        fprintf(stderr, "ERROR, no port provided\n");
        exit(1);
    }

    sockfd = socket(AF_INET, SOCK_STREAM, 0);
    if (sockfd < 0)
        error("ERROR opening socket");

    bzero((char *) &serv_addr, sizeof(serv_addr));
    portno = atoi(argv[1]);
    serv_addr.sin_family = AF_INET;
    serv_addr.sin_addr.s_addr = INADDR_ANY;
    serv_addr.sin_port = htons(portno);

    if (bind(sockfd, (struct sockaddr *) &serv_addr, sizeof(serv_addr)) < 0)
        error("ERROR on binding");

    listen(sockfd, 5);

    clilen = sizeof(cli_addr);
    newsockfd = accept(sockfd, (struct sockaddr *) &cli_addr, &clilen);
    if (newsockfd < 0)
        error("ERROR on accept");

    while (1) {
        bzero(buffer, 256);
        n = read(newsockfd, buffer, 255);
        if (n < 0)
            error("ERROR reading from socket");
```

```
        /* echo the same message back */
        n = write(newsockfd, buffer, strlen(buffer));
        if (n < 0)
            error("ERROR writing to socket");

        /* stop when the client types "Exit" */
        if (strcmp(buffer, "Bye", 3) == 0)
            break;
    }

    close(newsockfd);
    close(sockfd);
    return 0;
}
```

Client Code – exp2_client.c

```
/* Name-Sumit Mund
   date-24-08-2026
   sic - 24bcsh05
*/

/* Experiment 2 : TCP Echo Server that repeatedly echoes the message
   back to the client until the client types "Bye" -- CLIENT */

#include <stdio.h>
#include <sys/types.h>
#include <sys/socket.h>
#include <netinet/in.h>
#include <netdb.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>

void error(char *msg)
{
    perror(msg);
    exit(0);
}

int main(int argc, char *argv[])
{
    int sockfd, portno, n;
    struct sockaddr_in serv_addr;
    struct hostent *server;
    char buffer[256];

    if (argc < 3) {
        fprintf(stderr, "usage %s hostname port\n", argv[0]);
        exit(0);
    }

    portno = atoi(argv[2]);

    sockfd = socket(AF_INET, SOCK_STREAM, 0);
    if (sockfd < 0)
        error("ERROR opening socket");

    server = gethostbyname(argv[1]);
    if (server == NULL) {
        fprintf(stderr, "ERROR, no such host\n");
        exit(0);
    }
}
```

```
    bzero((char *) &serv_addr, sizeof(serv_addr));
    serv_addr.sin_family = AF_INET;
    bcopy((char *) server->h_addr, (char *) &serv_addr.sin_addr.s_addr, server-
>h_length);
    serv_addr.sin_port = htons(portno);

    if (connect(sockfd, (struct sockaddr *) &serv_addr, sizeof(serv_addr)) < 0)
        error("ERROR connecting");

    while (1) {
        printf("Please Enter the Message: ");
        bzero(buffer, 256);
        fgets(buffer, 255, stdin);

        n = write(sockfd, buffer, strlen(buffer));
        if (n < 0)
            error("ERROR writing to socket");

        bzero(buffer, 256);
        n = read(sockfd, buffer, 255);
        if (n < 0)
            error("ERROR reading from socket");

        printf("Echo from server: %s\n", buffer);

        /* exit when the user types "Bye" */
        if (strncmp(buffer, "Bye", 3) == 0)
            break;
    }

    close(sockfd);
    return 0;
}
```

Compile & Run Commands

Terminal 1: ./exp2_server 8200

Terminal 2: ./exp2_client localhost 8200

Sample Interaction

Input used: Client typed: Hello, Testing, Bye

Sample Output

Server Output:

Client Output:

Please Enter the Message: Echo from server: Hello

Please Enter the Message: Echo from server: Testing

Please Enter the Message: Echo from server: Bye

Experiment 3 - Chatting Application using TCP

Server Code – exp3_server.c

```
/* Name-Sumit Mund
   date-01-0-2026
   sic - 24bcsh05
*/

/* Experiment 3 : Develop a Chatting Application using TCP -- SERVER */

#include <stdio.h>
#include <stdlib.h>
#include <sys/types.h>
#include <sys/socket.h>
#include <netdb.h>
#include <signal.h>
#include <netinet/in.h>
#include <unistd.h>
#include <string.h>
#include <arpa/inet.h>
#define TRUE 1

int main()
{
    int soc, msgsock;
    socklen_t addrlen;
    char buf[1024], buf1[1024];
    struct sockaddr_in server, client;
    addrlen = sizeof(client);

    soc = socket(AF_INET, SOCK_STREAM, 0);
    if (soc < 0) {
        perror("Error opening socket");
        exit(1);
    }

    server.sin_family = AF_INET;
    server.sin_addr.s_addr = inet_addr("127.0.0.1");
    server.sin_port = 0;

    if (bind(soc, (struct sockaddr *) &server, sizeof(server)) < 0) {
        perror("Error binding");
        exit(1);
    }

    if (getsockname(soc, (struct sockaddr *) &server, &addrlen) < 0) {
        perror("Error getting port");
        exit(1);
    }
    printf("Socket has port no: %hd\n", ntohs(server.sin_port));

    listen(soc, 5);
    signal(SIGCHLD, SIG_IGN);

    do {
        msgsock = accept(soc, (struct sockaddr *) &client, &addrlen);
        if (msgsock == -1) {
            perror("Error in accept");
            exit(0);
        }

        if (fork() == 0) {
            close(soc);
            do {
```

```
        bzero(buf, 1024);
        read(msgsock, buf, 1024);
        printf("Message from client: %s\n", buf);

        printf("Message to client: ");
        bzero(buf1, 1024);
        fgets(buf1, 1024, stdin);
        write(msgsock, buf1, strlen(buf1));
    } while (strncmp(buf, "bye", 3) != 0);
    close(msgsock);
    exit(0);
}
close(msgsock);
} while (TRUE);

close(soc);
return 0;
}
```

Client Code – exp3_client.c

```
/* Name-Sumit Mund
   date-01-09-2026
   sic - 24bcsh05
*/

/* Experiment 4 : Develop a Chatting Application using TCP -- CLIENT */

#include <stdio.h>
#include <stdlib.h>
#include <sys/types.h>
#include <sys/socket.h>
#include <netdb.h>
#include <signal.h>
#include <netinet/in.h>
#include <unistd.h>
#include <string.h>
#include <arpa/inet.h>

int main(int argc, char *argv[])
{
    int soc;
    char buf[1024], buf1[1024];
    struct sockaddr_in server;
    socklen_t addrlen = sizeof(server);

    if (argc < 2) {
        fprintf(stderr, "usage %s port\n", argv[0]);
        exit(1);
    }

    soc = socket(AF_INET, SOCK_STREAM, 0);
    if (soc < 0) {
        perror("Error opening socket");
        exit(1);
    }

    server.sin_family = AF_INET;
    server.sin_addr.s_addr = inet_addr("127.0.0.1");
    server.sin_port = htons(atoi(argv[1]));

    if (connect(soc, (struct sockaddr *) &server, sizeof(server)) < 0) {
        perror("Error in connection");
    }
}
```



```
        exit(1);
    }

    do {
        printf("Client input: ");
        bzero(buf, 1024);
        fgets(buf, 1024, stdin);
        write(soc, buf, strlen(buf));

        bzero(buf1, 1024);
        read(soc, buf1, 1024);
        printf("Message from server: %s\n", buf1);
    } while (strncmp(buf, "bye", 3) != 0);

    close(soc);
    return 0;
}
```

Compile & Run Commands

Terminal 1: `./exp3_server` (prints the port, e.g. 34633)
Terminal 2: `./exp3_client 34633`

Sample Interaction

Input used: Client: Hello / How have you been? / bye
Server: Hii, welcome to chat! / I am good, thanks! / bye

Sample Output

Server Output:

```
Socket has port no: -30903
Message from client: Hello

Message to client: Message from client: How have you been?

Message to client: Message from client: bye

Message to client:
```

Client Output:

```
Client input: Message from server: Hii, welcome to chat!

Client input: Message from server: I am good, thanks!

Client input: Message from server: bye
```